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What Makes Social Media so Addictive, and How Does it Affect our Mental Health?

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Abstract

What Makes Social Media so Addictive, and How Does it Affect our Mental Health?

Hey there! Have you ever mindlessly scrolled through your social media feed, losing track of time? Social media platforms have become a huge part of many people's daily lives, but researchers are becoming more worried about how addictive they can be. The term "social media addiction" describes behaviors similar to those seen in substance use disorders, like the compulsive need to check notifications and craving likes and comments (Turel et al., 2014, p. 678). Isn't it interesting how a simple "like" can impact our mood? This paper will dig into what makes social media so addicting, focusing on factors like dopamine release, instant gratification, habit formation, and the need for social validation, along with how these things negatively affect mental health.

Role of Dopamine

One big reason social media is addictive has to do with something called dopamine, which is a chemical in our brain related to the reward system. Neuroimaging studies show that brain regions light up when people use social media. This happens in the part of the brain responsible for processing rewards. Turel et al. (2014) did this type of research on the effects of social media, neural systems in the brain, and addiction. The research included 65 young adults aged 18 to 23 recruited from a university. The participants were checked for how much they used Facebook and filled out a questionnaire to measure signs of Facebook addiction. Based on their scores, they were grouped into low to medium levels of addiction-like symptoms. The researchers used tasks to measure how quickly the participants responded and how their brains reacted to Facebook-related images compared to neutral ones (Turel et al.,2014, pp. 678-679). Guess what? The results showed that those little notifications triggered a dopamine rush and played a big role in Facebook addiction. When participants had higher addiction scores, there was more activity in the brain's reward system area. This means their brains reacted strongly to

Facebook cues. Overall, the symptoms of addiction seemed to come from strong connections in their brains related to Facebook, emphasizing how dopamine influences addictive behavior on social media (Turel et al., 2014, pp. 684 - 687). The release of dopamine creates a cycle of wanting and reward that drives users to continue to the platform in a repetitive, addictive way. This may remind you of a system of gold stars that you may have received in school for good behavior or achieving specific goals. Each like or positive comment on social media is like earning a gold star. This provides a boost, which is a reward that encourages us to return for more. Does this make you wonder what specific features of Facebook and other social media, like notifications or "likes," might trigger dopamine responses in users? Do you think you feel good when you are doing things on a social media platform like Facebook?

Instant Gratification

Another big reason social media is addictive is the instant gratification factor. Social media platforms are designed so that users are provided with instant gratification, which means that when someone wants something, they expect to receive it quickly. Because this is how social media platforms are designed, it is tough to resist. According to Flanner et al. (2024), "Social feedback is both frequent and quantifiable (e.g., number of likes, number of followers) via social media. Such social feedback is also usually reinforcing and delivered on a variable ratio reinforcement schedule which is highly resistant to behavior extinction and is thus particularly addictive…" (Flannery et al., 2024, p. 1). This immediate availability of rewards drives compulsive checking behaviors. In other words, instant gratification is satisfying, at least for a while. According to Montag et al. (2019), social media unpredictably provides rewards, but people still seek instant gratification because sometimes they get it. This cycle leads to compulsive behaviors (Montag et al., 2019, p. 5). This unpredictable or intermittent

reinforcement is very similar to slot machines, which also somehow keep users engaged. People continue to pull the lever, thinking they will hit it big and win on the next pull. When they do win, they are excited with instant gratification. When they do not win, they keep going, thinking they will win on the next pull. When they do win, they continue to play because they think they can win an even bigger amount on another pull. This comparison highlights how the reinforcement of social media works. Additional research has been done by Ryan et al. (2014), who reviewed many studies about Facebook addiction and concluded that the system of likes, comments, and shares provides users with immediate rewards, which reinforce the behavior and make the platform difficult to leave" (Ryan et al., 2014, p. 136). How does the comparison between social media rewards and playing slot machines help us understand the addictive nature of social media? Have you ever caught yourself checking your notifications during a conversation? This raises the question: How can individuals manage their social media use and reduce reliance on instant gratification?

Habit Formation

Another piece of the puzzle of social media addiction is that social media platforms are also very effective at forming habits that can lead to addiction over time. An article by Montag et al. (2019) analyzed how smartphones and Internet apps keep people using them for longer. The authors look at past studies and data about smartphone use, social media, and their effects on people's minds, including things like comparing ourselves to others, addiction, and how engaged users feel. Combining information from different studies shows how specific app features affect people's behavior and feelings (Montag et al., 2019, p. 4). In this article, Montag et al. (2019) suggested that social media platforms like YouTube use endless scrolling and autoplay features to make people stay on longer. As users scroll, they keep finding new and interesting content,

which makes them want to keep going. Since the feed is endless, the user does not think about ending and going to another page or stopping. It is similar to how slot machines work, offering rewards at random times to keep you hooked. These features encourage users to form habits, making stopping and leaving the platform harder (Montag et al., 2019, p. 5). Furthermore, "... app developers have a keen interest in designing online platforms with the goal to keep users occupied as long as possible" (Montag et al., 2019, p. 4). This is because a business model used in Silicon Valley does not charge a user money but charges in a way with personal data instead. All of this promotes habit formation and addiction for users of social media. These repeated actions of checking social media form a habit that is difficult to break. In addition, the research using neuroimaging showed that individuals with a Facebook addiction demonstrated decreased activity in various regions of the brain that are responsible for self-control. The more frequently an individual engages in these habitual behaviors, the more significant the loss of self-control (Turel et al., 2014, pp. 684-687). This also makes it harder to break the Facebook habit since the individual involved has less control, and that control gets weaker the more they engage in the site. Reflect on your own involvement with social media platforms: Have you noticed that once you start checking a site regularly, you seem to check more and more? Do you feel like your usage of social media has become a habit? Do you feel a sense of being controlled by it?

Social Validation

Another major factor contributing to social media addiction is the constant need for social validation. Many social media platforms provide instant feedback through likes, comments, and shares. These allow users to gain social validation from others for their posts or connections with various social media content. After looking at many studies about Facebook addiction, Ryan et al. concluded that in most cases, "...passively engaging with social or entertainment related

content on Facebook can motivate users to return to the site frequently. … Such behavior may be tapping into what is anecdotally referred to as fear of missing out or FoMO" (Ryan et al., 2014, p. 139). Other research supports this as well. According to Montag et al. (2019), "Likes demonstrate either positive social feedback on one's own posts or to give another person such a feedback" (Montag et al., 2019, p. 6). Both types of positive feedback are a big part of social validation. The power of receiving likes on social media is backed by neuroscience. Studies have shown that when users see their posts with many likes, their brains respond by activating areas linked to rewards, like the ventral striatum. This creates a feeling of pleasure, reinforcing their desire to keep using the platform. It is similar to how rewards work in other parts of life, making the act of getting likes addictive and influencing people to spend more time on social media (Montag et al., 2019, p. 6). Consider your social media use: Do you check how many likes your post gets? Do you feel disappointed when you do not get as many likes as you expected?

Negative Effects on Mental Health

It is unbelievable how we have created a culture where we might gauge our self worth based on how many likes a photo gets. Does that make you think twice about the pressure we put on ourselves with social media? So, this is the insane part, while social media can connect us, excessive use can seriously harm our mental health. A study by Flannery et al. (2024) looked at 150 teenagers (75 girls and 75 boys) to explore how their brains reacted to social feedback and how this relates to addiction-like social media use. The researchers used brain scans to check activity in areas of the brain that deal with social feedback. They wanted to see how these brain responses changed as teens developed and how this was connected to feelings of depression, especially looking at differences between boys and girls. Their results showed that girls are more affected by addictive social media use, which could lead to depression due to their higher levels

of social media addiction (Flannery et al., 2024, pp. 2-7). In other research mentioned earlier by Montag et al. (2019), one of the findings was that social media use may cause "...withdrawal from society and escapism to the online world ultimately might cause loneliness and reduce social connectedness." (Montag et al., 2019, p. 8). So even though the research by Flannery et al. (2024) suggested social media impacts girls' mental health more than boys, Montag et al. research indicates that it can affect anyone who uses social media. In addition, Pontes, another researcher, used a survey to collect information from 495 adolescents which included 253 males and 242 females, about their experiences with social networking site addiction and Internet gaming disorder, as well as their mental health, including symptoms of depression, anxiety, and stress. Pontes found that both social networking site addiction and Internet gaming disorder have been found to negatively impact psychological health, leading to heightened levels of depression, anxiety, and stress (Pontes, 2017, p. 609). So, there is evidence that social media can affect our mental health. Does this make you wonder if your need for social connection online is more important than the potential damage to your mental health and well being? Have you ever felt more isolated after having spent time online?

Conclusion

So, what is the takeaway? Social media addiction is a complex issue with many parts, including the role of dopamine, instant gratification, habit formation, and social validation. All of these pieces are designed to keep us engaged and connected, but is the cost of our mental health worth it? As you navigate this world of social media usage, you need to find a balance between staying connected and taking care of your mental health. What small changes can you make in your social media usage to make sure it enhances your life rather than detracts from it?

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